

ENTERPRISE ARCHITECTURE (EA) AND DATABASE PROGRAMMING

Azadeh Sarkheyli, Ph.D.
(azadeh.sarkheyli@hh.se)

CONTENTS:

- Course plan (Weeks 18-26)
- Introduction to Enterprise Architecture (EA)
- Project Proposal

COURSE PLAN

Learning Outcomes (LOs)

After completing the course, the student should be able to:

Knowledge and understanding

- Explain the concepts of architectural strategy, system structuring and Enterprise Architecture (EA).
- Describe principles for a service-oriented, component-based, and distributed architecture.
- Describe relationships between different types of architecture within Enterprise Architecture (EA): business architecture, information architecture and IT architecture.
- Demonstrate understanding of different roles in Enterprise Architecture (EA).

Skills and Abilities

- Use general theory, methodology and technology for process and data modeling.
- Analyze a business and based on the analysis, design a database architecture that corresponds to a business's needs, goals, and requirements.
- Implement different database solutions to achieve optimal database performance.
- Orally and in writing present problems and solutions in the area.

Evaluation

- Assess whether different database solutions are reusable and optimized based on criteria for good database design.
- Based on relevant articles, discuss, reflect, and argue for when a certain type of architecture may be suitable, and for which type of business.

Week	Date	Type (Hours)	Content	Reading	Zoom Link
18	01 May	Lecture1	- Course information and introduction	Lankhorst's book ch1	Online * Zoom Link
19	08 May	Seminar1*GA	- Project proposal		
20	14 May	Seminar1*GB	- Project proposal		
21	21 May	Lecture2	- Introduction to EA	Lankhorst's book ch1 &2	
22	28 May	Seminar2*GA	- Modelling and Analysis		
23	01 June	Seminar2*GB	- Modelling and Analysis		JUFE Campus
	02 June	Lecture3	- EA and Other Governance Instruments	Lankhorst's book ch 2	
	03 June	Seminar3*GA	- Modelling and Analysis		
	04 June	Seminar3*GB	- Modelling and Analysis		
	05 June	Lecture4	- Architecture Methods and Frameworks - Description Languages	Lankhorst's book ch2	
24	08 June	Seminar4*GA	- Modelling and Analysis		
	09 June	Seminar4*GB	- Modelling and Analysis		
	10 June	Lecture5	- Foundations	Lankhorst's book ch3	
	12 June	Seminar5*GA	- Modelling and Analysis		
25	15 June	Seminar5*GB	- Modelling and Analysis		Online * Zoom Link
	17 June	Lecture6	- Communication of Enterprise Architectures - Enterprise Modelling	Lankhorst's book ch4&5	
	18 June	Seminar6*GA&B	- Modelling and Analysis		
26	22 June	Seminar7*GA&B	- Final Seminar		
26	23 June	Seminar7*GA& B	- Final Seminar		
	26 June	Written Test			

* Seminars will be held for two main groups (A & B)

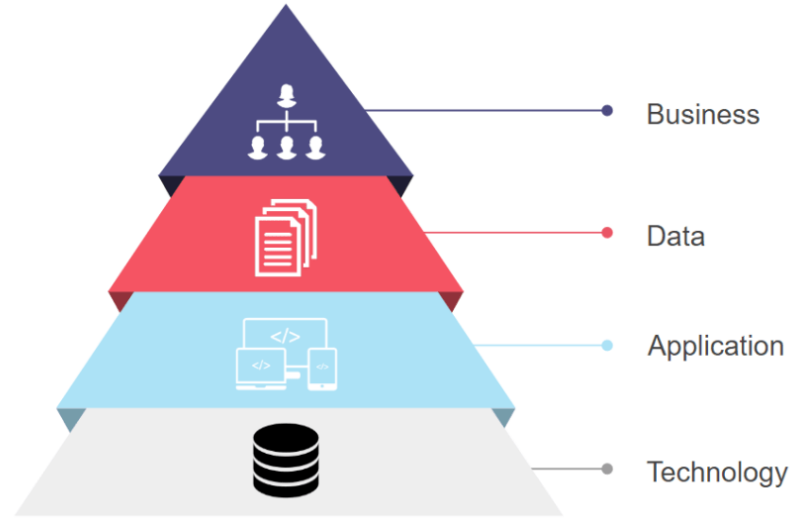
* [Zoom link](#) is valid for all the sessions

COURSE PLAN

COURSE PLAN

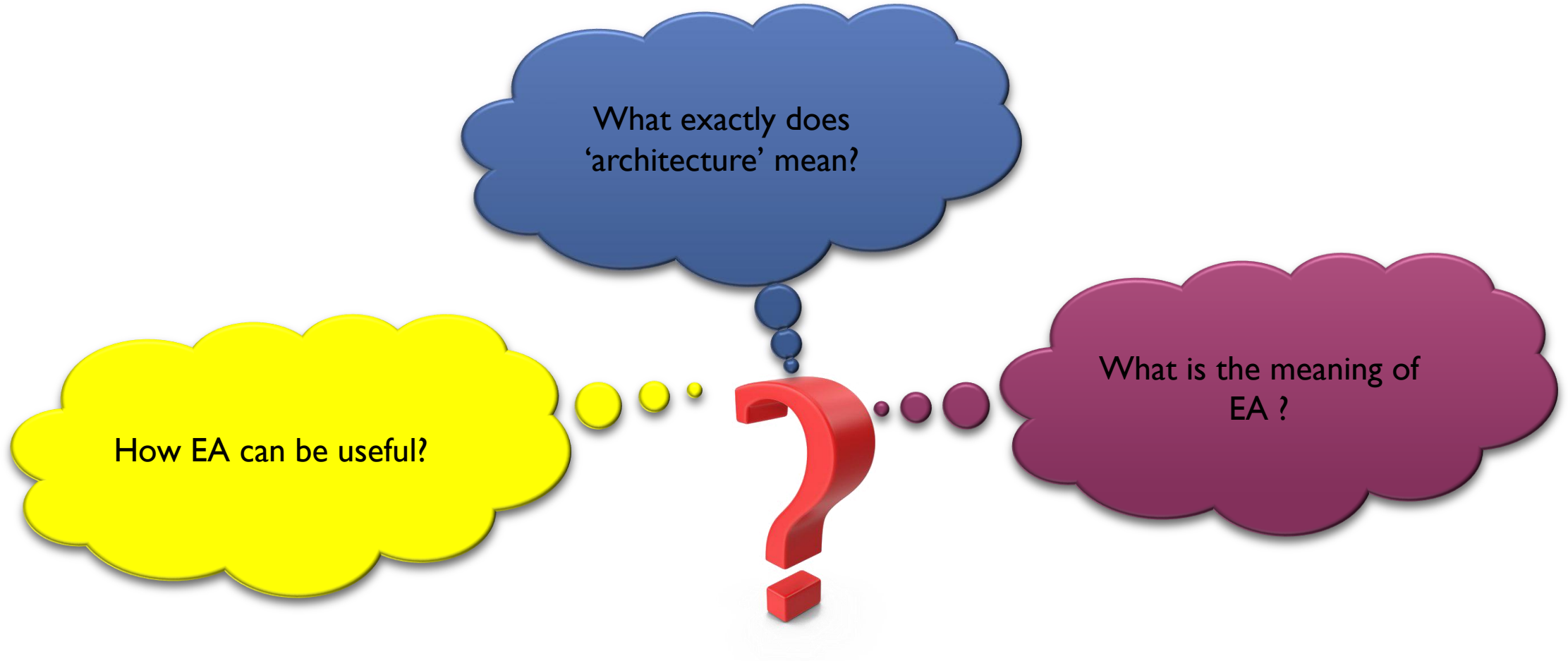
Assessment

- Assessment 1: Attendance, cooperation, oral presentation in seminars, (30% of the overall grade)
- Assessment 2: Group project work with a written report and oral presentation, (30% of the overall grade)
- Assessment 3: Individual written test, (40% of the overall grade)



INTRODUCTION TO ENTERPRISE ARCHITECTURE

INTRODUCTION



INTRODUCTION: CONCEPTS

- To manage the complexity of any large organization or system, you need **architecture**.
- For example, an architectural model of a house defines its major functions and how they are structured. It provides an abstract design, ignoring many details.
- Moreover, a similar frame of reference is needed in designing an enterprise. To create an overview of the structure of an organization, its business processes, their application support, and the technical infrastructure, you need to express the **different aspects and domains, and their relations**.
- **Architecture** can signify the art and science of designing the built environment, or the product of such a design. Thus, the term architecture can encompass both the blueprint for a building and the general underlying principles such as its style, as in 'gothic architecture'.



INTRODUCTION: CONCEPTS

Architecture: fundamental concepts or properties of a system in its environment, embodied in its elements, relationships, and in the principles of its design and evolution.



INTRODUCTION

Discussion

- What is the meaning of **stakeholder**? (with example)
- Why should we consider it?
- What is the meaning of **enterprise**?



INTRODUCTION: HISTORICAL BACKGROUND OF ENTERPRISE ARCHITECTURE

Roger Sessions, Object Watch Inc (2007):

- Thirty years ago, a new field was born that soon came to be known as enterprise architecture. The field initially began to address two problems:
 - **System complexity**—Organizations were spending more and more money building IT systems; and
 - **Poor business alignment**—Organizations were finding it more and more difficult to keep those increasingly expensive IT systems aligned with business need
- IT systems unmanageably complex and increasingly costly to maintain
- IT systems hindering the organization's ability to respond to current, and future, market conditions in a timely and cost-effective manner
- Mission-critical information consistently out-of-date and/or just plain wrong
- A culture of distrust between the business and technology sides of the organization

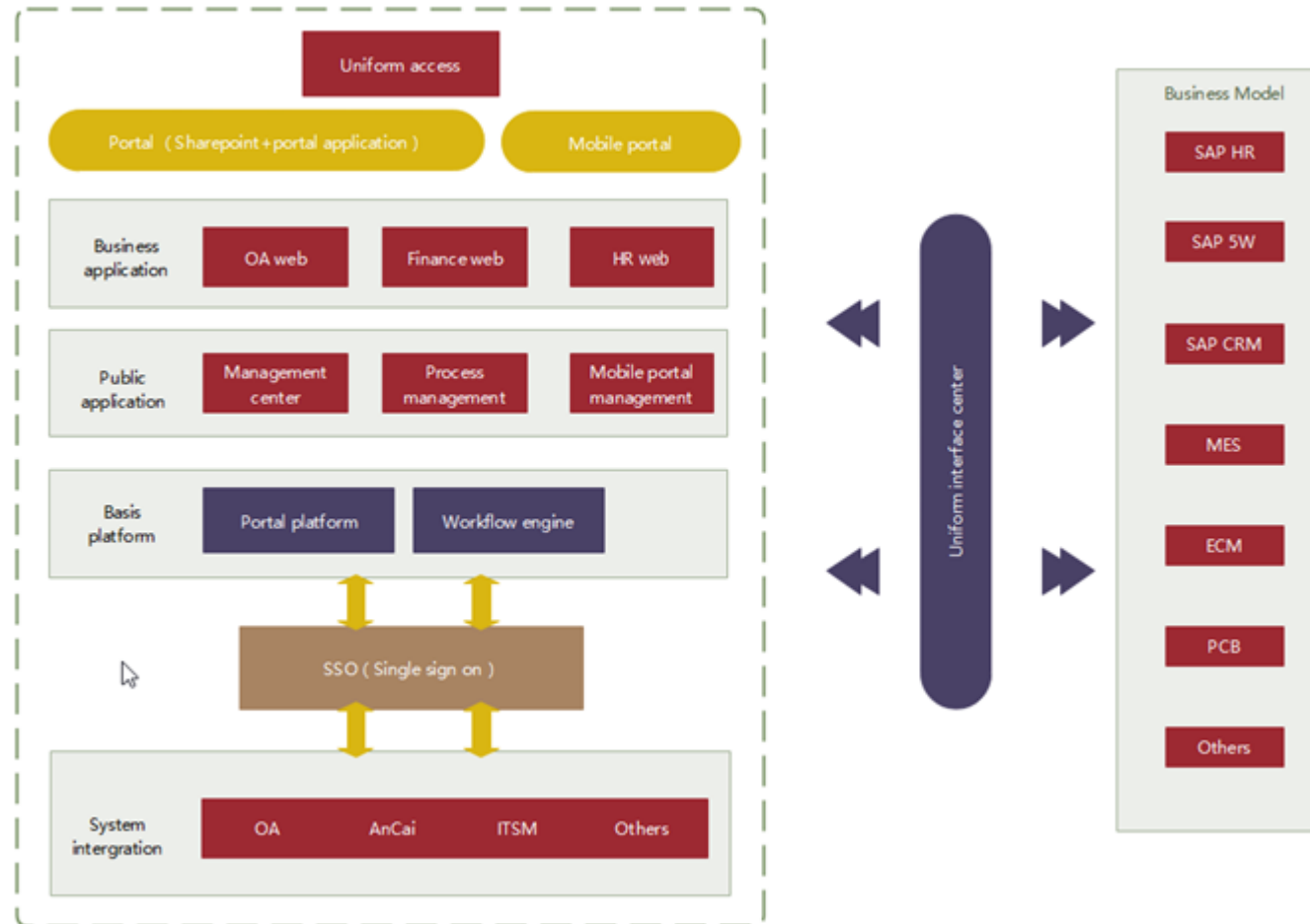
INTRODUCTION: DEFINITIONS

- **Enterprise:** any collection of organizations that has a common set of goals and/or a single bottom line.
- **Enterprise architecture:** a coherent whole of principles, methods, and models that are used in the design and realization of an enterprise's organizational structure, business processes, information systems, and infrastructure.
- ... the organizing logic for business processes and IT infrastructure reflecting the integration and standardization requirements of the company's operating model ... the desired state of business process integration and business process standardization for delivering goods and services to customers (Peter Weill, MIT Center for IS Research)
- ... defines the mission of an agency and describes the technology and information needed to perform that mission, along with descriptions of how the architecture of the organization should be changed in order to respond to changes in the mission (United States Government)
- ... the process of translating business vision and strategy into effective enterprise change by creating, communicating and improving the key requirements, principles and models that describe the enterprise's future state and enable its evolution (Gartner)

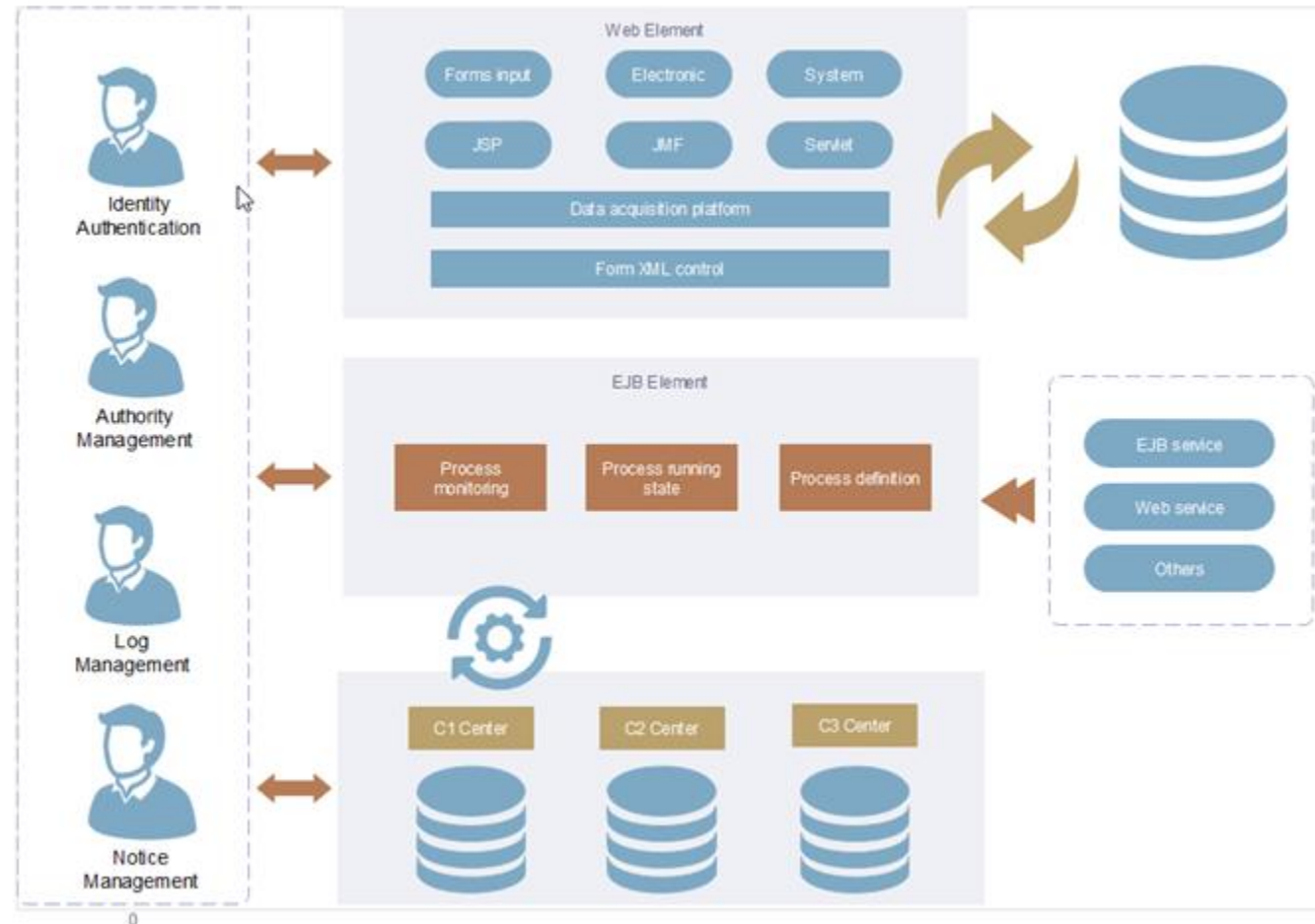
INTRODUCTION: ENTERPRISE ARCHITECTURE SCOPE

- The term enterprise is generally applicable in many circumstances, including
 - Public or private sector organizations
 - An entire business or corporation
 - A part of a larger enterprise (such as a business unit)
 - A conglomerate of several organizations, such as a joint venture or partnership
 - A multiply outsourced business operation
- The term enterprise includes the whole complex, socio-technical system, including:
 - people
 - information
 - technology
 - business (e.g., operations)
- Defining the boundary or scope of the enterprise to be described is an important first step in creating the enterprise architecture. “Enterprise” means more than the information systems employed by an organization.
- Composing holistic solutions that address the business challenges of the enterprise and support the governance needed to implement them (Gartner).

INTRODUCTION: ENTERPRISE ARCHITECTURE EXAMPLE



INTRODUCTION: ENTERPRISE ARCHITECTURE EXAMPLE



INTRODUCTION

Discussion

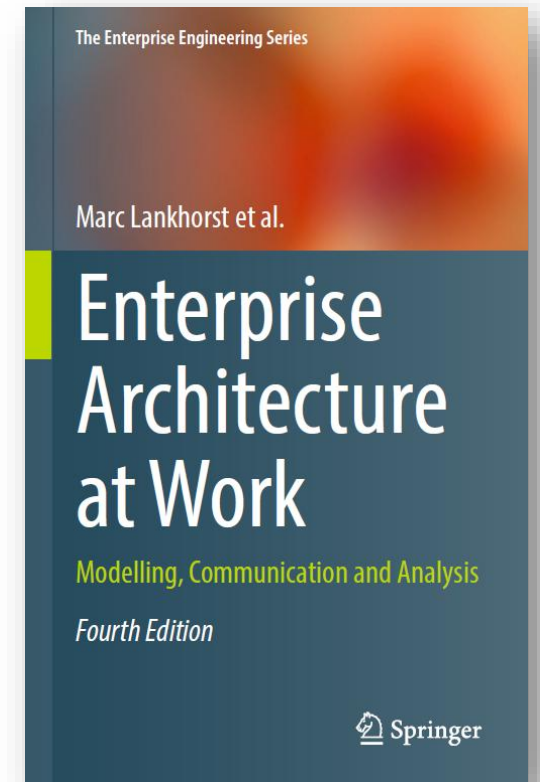
According the below papers describe **why do we need Enterprise Architecture?**

- ❖ Shanks, G., Gloet, M., Someh, I.A., Frampton, K., & Tamm, T. (2018). Achieving benefits with enterprise architecture. The Journal of Strategic Information Systems, 27(2), 139-156.
- ❖ van de Wetering, R., Kurnia, S., & Kotusev, S. (2021). The role of enterprise architecture for digital transformations. Sustainability, 13(4), 2237.
- ❖ Ross, J.W. (2006). Enterprise architecture: Driving business benefits from IT.



REFERENCES

- Lankhorst, M. (2017). Enterprise architecture at work (ISBN 978-3-662-53933-0). Berlin: Springer.
- <https://loggle.io/blog/5-benefits-of-enterprise-architecture>
- <https://www.edrawsoft.com/enterprisearchitecture/enterprise-architecture-examples.html>



PROJECT

Form your group

- The students should form a group consisting of 4 or 5 members.

